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Received: 3 July 2023 | Revised: 28 July 2023 | Accepted: 8 August 2023

DOI: 10.1111/ctr.15101

Clinical TRANSPLANTATION
The Journal of Clinical and Translational Research
WILEY

ORIGINAL ARTICLE

Long-Term outcomes in adult patients with congenital heart disease considered for transplantation: A single center study

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Funding information
National Institutes of Health, Grant/Award Number: K08 HL 161440; American Heart Association, Grant/Award Number: 852875; HeartFest

Abstract

Background: Adult congenital heart disease (ACHD) patients pose unique challenges in identifying the time for transplantation and factors influencing outcomes.

Objective: To identify hemodynamic, functional, and laboratory parameters that correlate with 1- and 10-year outcomes in ACHD patients considered for transplantation.

Methods: A retrospective chart review of long-term outcomes in adult patients with congenital heart disease (CHD) evaluated for heart or heart + additional organ transplant between 2004 and 2014 at our center was performed. A machine learning decision tree model was used to evaluate multiple clinical parameters correlating with 1- and 10-year survival.

Results: We identified 58 patients meeting criteria. D-transposition of the great arteries (D-TGA) with atrial switch operation (20.7%), tetralogy of Fallot/pulmonary atresia (15.5%), and tricuspid atresia (13.8%) were the most common diagnosis for transplant. Single ventricle patients were most likely to be listed for transplantation (39.8% of evaluated patients). Among a comprehensive list of clinical factors, invasive hemodynamic parameters (pulmonary capillary wedge pressure (PCWP), systemic vascular pressure (SVP), and end diastolic pressures (EDP) most correlated with 1- and 10-year outcomes. Transplanted patients with SVP < 14 and non- transplanted patients with PCWP < 15 had 100% survival 1-year post-transplantation.

Conclusion: For the first time, our study identifies that hemodynamic parameters most strongly correlate with 1- and 10-year outcomes in ACHD patients considered for transplantation, using a data-driven machine learning model.

KEYWORDS
classification systems, clinical decision-making, heart disease: congenital

1 | INTRODUCTION

Most patients with congenital heart disease (CHD) have successfully survived through childhood due to advances in pediatric surgical management during this past era.[1] As the fastest growing population of adults with heart disease,[2] adult CHD patients have introduced many challenges in decision making, especially when they are evaluated for life prolonging treatments. Adults with CHD who decline in functional status and need a heart/heart+ additional organ(s) (heart/heart+) transplant are distinct from traditional patients who were born with

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structurally normal hearts but develop ischemic or non-ischemic cardiomyopathy requiring evaluation for transplantation. While most traditional patients with heart failure benefit from guideline directed medical therapy, cardiac resynchronization therapy and can be bridged with mechanical assistance devices, the benefit of these therapies in adult congenital heart disease (ACHD) patients is uncertain and the patient's anatomy often prohibits their use.[3,4] Varied cardiac and vascular anatomic presentations result in hemodynamic variations with little commonality to identify prognostic factors which may influence outcomes in an already numerically small patient population. The pediatric CHD patient undergoes numerous interventions in childhood, and the burden of these interventions in the form of sternotomies, myocardial scarring, and allosensitization[5] adds to technical complexity of the ACHD patients when being considered for transplantation. Most CHD patients are considered for organ transplantation when other interventions fail and are significantly frail when listed for transplantation. Consequently, these patients either die while waiting for organs or shortly after transplantation, either due to technical issues or poor resilience.[6] Hence, it is important to identify clinical parameters that can predict optimal time for transplantation in ACHD patients.

To date, studies that investigate transplant outcomes in ACHD patients come mostly from organ transplantation registries that are limited in patient specific clinical information. Factors that drive long-term survival in ACHD patients are unknown, contributing to limitations in identifying patients and timing for transplantation in this high-risk population. In this paper, we describe a single-center retrospective cross sectional study involving 58 ACHD patients that evaluates a comprehensive set of hemodynamic, functional, and laboratory parameters used to determine candidacy for transplantation and identifies factors that were related to 1- and 10-year survival. Our study, for the first time, shows that pre-listing hemodynamic parameters most strongly correlate with survival in ACHD patients.

2 | METHODS

2.1 | Study population

Using electronic medical records, we performed a comprehensive retrospective analysis of all ACHD patients ≥18 years, referred for heart and heart+ transplantation from November 2004 to July 2014 at University of Pittsburgh Medical Center (UPMC). Seventy-one patients evaluated for heart/heart+ transplant were identified. Of these, 58 patients had all relevant data (laboratory tests, cardiopulmonary exercise test [CPET], invasive hemodynamic testing, imaging studies) available for evaluation and were included in the study. The 58 patients were evaluated for clinical outcomes until June 2021 (Table S1).

2.2 | Transplant assessment

Evaluation for transplant candidacy was conducted by a multidisciplinary team consisting of ACHD specialists, heart failure-transplant specialists, cardiothoracic surgeons including those specializing in CHD surgery, transplant coordinators, nurses, and liver or pulmonary transplant specialist when indicated. Following evaluation, patients were divided into one of three categories, (1) listed for transplantation, (2) deferred or "too well" for transplantation, and (3) declined or "too sick" for transplantation. Patients accepted for transplantation were grouped as either "waitlisted" but not transplanted (waitlisted) or transplanted based on their transplant status at the end of the study.

2.3 | Chart review

Chart review comprised of a comprehensive collection of demographic information, CHD diagnosis, subsequent surgeries including number of sternotomies and subjective functional evaluation including New York Heart Association (NYHA) Class at the time of evaluation. Objective functional data including CPET parameters were collected. Hemodynamic data collected from right heart catheterization and laboratory data at the time of decision for listing were also collected. Outcomes including mortality after transplantation as well as mortality after decision for transplant candidacy were collected. Catheter based hemodynamic testing, laboratory and functional (exercise testing) criteria were evaluated to determine factors that correlated with decision to listing ACHD patients for transplant. Glenn pressures were noted as systemic venous pressure (SVP) in patients with Fontan physiology. Also, pulmonary artery (PA) pressures were derived from an average of left PA (LPA) and right PA (RPA) pressures in Fontan patients.

2.4 | Statistics

Statistical analysis was conducted in JMP, Pro 15.2. SAS Institute Inc., Cary, NC, 1989–2021. Normality of data was analyzed using the Kolmogorov–Smirnov test. Significance was defined as a type I error rate of 5% ($p < .05$) for all univariate analysis. Univariate analysis of categorical variables was conducted using logistic regression with likelihood ratio chi-square testing. Parameter estimate odd ratios were calculated for each variable. Kruskal–Wallis nonparametric analysis was conducted to compare differences between variables in patients listed for transplant (transplanted + waitlisted) versus not listed (too sick + too well). Contingency analysis using Pearson's chi-square testing was done to assess the relationship between single ventricle physiology and outcome. To identify variables predicting the categorical outcome of 1- and 10-year mortality as well as relevant cut-off values we used machine learning decision-tree analysis in JMP. We considered all previously significant variables from univariate analysis into our multivariate model, including listing decision, advanced support, ACHD complexity, SVP, wedge pressure, and sternotomies, dichotomized into two groups. Waitlisted patients who did not receive a transplant and patients who were too sick for transplantation (declined) were grouped into a category for analysis as they both were deemed to need transplantation and could not get it. Deferred patients were not evaluated for 1-year outcome using this model. For

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10-year outcome analysis, transplanted and patients who were too well (deferred) were similarly grouped into a category as they both had competent organs at the end of the evaluation period. At each step, partition analysis chooses the independent (predictor) variable that has the strongest interaction with the dependent variable. This variable is split to maximize the difference in responses between the two nodes of the split. This analysis then creates a model which maximizes the R^2 value, or the measure of the amount of variation explained by splitting the data into groups. We then verified this model against a subsection of our existing dataset using K-fold cross validation Kaplan–Meier plot was created to analyze survival between our four decision groups.

3 | RESULTS

3.1 | Systemic right ventricle was the most common anatomic diagnosis for transplantation

The patients' baseline characteristics are listed in Table 1. Patients had a mean age of 35.8 years [29.3–44 years], with 65.5% being male. 91.4% were Caucasian, 3.4% were Black, and 5.2% were Asian which is typical demographic distribution for ACHD patients considered for transplantation across the country. While 2 (3.4%) were NYHA Class II, majority of the patients were reported as NYHA Class III (62.1%) and Class IV (34.5%). The most common CHD diagnosis considered for transplantation was D-transposition of the great arteries (D-TGA) (20.7%) (with atrial switch), followed by Tetralogy of Fallot/pulmonary atresia (15.5%) and tricuspid atresia with Fontan palliation (13.8%). Single ventricle physiology with or without Fontan palliation represented 19 (32.8%) of patients. Age at evaluation, sex, race, NYHA class, and CHD diagnosis were not associated with the decision for transplant listing. Multiple comorbidities were evaluated (Table 1) and were not associated with the listing decision. However, single ventricle physiology patients were the most likely to be listed for transplant and accounted for 35.7% of the listed patients.

3.2 | Invasive hemodynamic testing and functional exercise data are associated with listing decisions

Patients were considered for transplantation after a comprehensive evaluation by a multidisciplinary team. Criteria used to evaluate eligibility for transplantation in non-ACHD patients, such as medication compliance, financial and social eligibility, PRA sensitization, and BMI > 35 kg/m^2, and other standard criteria (ongoing malignancy, medical comorbidities that would preclude transplantation, etc.) were used to screen patients. Number of sternotomies ($p = .0012$) and systolic blood pressures ($p = .0178$) were - associated with listing status. SVP (14.5 vs. 10 vs. 23 mmHg, $p = .0052$) and systemic ventricular end-diastolic pressures (SVEDP)(15 vs. 12 vs. 25 mmHg, $p = .0132$) also correlated with decision to list. Peak VO2 (15 vs. 24 vs. 11.7 kg/min/m^2, $p = .0354$) and FEV1/FVC (80 vs. 68 vs. 71, $p = .0136$) from CPET associated with listing for transplant versus assigning a deferred or declined status, respectively. Five ACHD patients were on left ventricular assist devices (LVAD) during decision for transplantation and were all listed for transplantation. While the number of patients on advanced support therapies (inotropes, LVAD, extracorporeal membrane oxygenation [ECMO]) were too small for statistical analysis, 16 (76.2%) patients on inotropes were listed for transplantation whereas 4 (19%) were deemed to be too sick for transplantation. One patient on inotropes and ECMO each were deferred as they had non-cardiac causes exacerbating their clinical condition. Factors not associated with decision-making included oxygen saturation (an indicator of cyanotic disease), and parameters derived from CPET including Max HR, minute ventilation/carbon dioxide production (VE/VCO2) slope, DLCO. Exercise time was not considered for analysis as different exercise protocols were used during testing. Laboratory tests including albumin, creatinine, serum sodium, and BNP did not correlate with listing decision. Other values calculated during catheter-based hemodynamic assessment including PVR, CO, CI, and PA pressure were not associated with listing decision. Complexity of the CHD itself as determined by anatomic class did not correlate with listing decision; however, number of sternotomies served as a better predictor of technical complexity (Table 2).

3.3 | Transplantation and invasive hemodynamic parameters correlate with 1-year survival

Transplantation was the strongest predictor of 1 year survival ($p < .0022$) (Figure 1A). Transplanted patients had a 3.42 times likelihood of survival at 1 year compared to patients declined for being too sick ($p = .067$). Patients deferred for transplantation (too well) had 100% 1-year survival, validating our screening process for transplanting patients prematurely. Catheter based hemodynamic testing demonstrated that deferred patients had a mean wedge pressure = 12 mmHg, SVP = 10 mmHg, and had a max myocardial oxygen consumption (MVO2) = 24 mL/kg/min on CPET. Waitlisted patients who did not receive a transplant were 5.15 times more likely to die at 1-year compared to transplanted patients ($p = .0047$). Patients listed for multi-organ transplantation had similar outcomes as patient listed for heart-transplant alone. Multivariate decision tree analysis (Figure 1B) showed that transplanted patients with pre-listing-decision SVP < 14 mmHg had 100% survival at 1 year while patients with SVP > 14 mmHg had a 66.7% survival. Number of sternotomies, a surrogate of CHD complexity, did correlate with survival counterintuitively; patients with ≤2 median sternotomies who waitlisted/declined had an 83.3% mortality at 1 year. However, the patients who were waitlisted (but not transplanted at 1 year) or declined for transplantation with >2 sternotomies and pulmonary capillary wedge pressures (PCWPs) <15 mmHg had a 100% 1-year survival. Declined/waitlisted patients with ≤2 sternotomies (TGA with systemic right ventricle [cc-TGA or D-TGA with atrial switch]: 7, TOF: 2, single ventricle: 5, uncorrected VSD1, Ebstein anomaly: 1) and on advanced support died, likely representing the sickest patient.

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TABLE 1 Demographic characteristics of adult patients with congenital heart disease listed for transplant.

	Deferred/too well *N* (%)	Listed for transplant *N* (%)	Declined/too sick *N* (%)	*p*-value
N	5 (8.6)	42 (72.4)	11 (18.9)	
Age (years) [25th, 75th]	40 [25.5, 47]	34 [26.8, 42.5]	40 [31, 50]	.512
Female sex *n* (%)	1 (20)	15 (35.7)	4 (36.4)	.758
Race (%)				.649
White	5 (100)	39 (92.9)	9 (82)	
Black	0 (0)	1 (2.4)	1 (9)	
Asian	0 (0)	2 (4.8)	1 (9)	
BMI [25th, 75th] (kg/m^2)	25.9 [22.5, 33.7]	24 [21, 27]	24.6 [21, 26]	
NYHA FC (%)				.749
II	1 (2.4)	0 (0)	1 (9.1)	
III	4 (80)	26 (61.9)	6 (54.5)	
IV	1 (20)	15 (35.7)	4 (36.4)	
CHD diagnosis (%)				.793
D-transposition of the great arteries with atrial switch (Senning or Mustard)	1 (20)	10 (23.8)	1 (9.1)	
TOF/pulmonary atresia	1 (0)	5 (7.1)	3 (18.2)	
Tricuspid atresia	0 (0)	6 (14.3)	2 (18.2)	
L-transposition of the great arteries	1 (20)	6 (14.3)	0 (0)	
Double inlet left ventricle	0 (0)	4 (9.5)	0 (0)	
Ventricular septal defect	1 (20)	1 (2.4)	1 (9.1)	
Shone complex	0 (0)	1 (2.4)	0 (0)	
Truncus arteriosus	0 (0)	0 (0)	1 (9.1)	
Hypoplastic left heart	1 (20)	1 (2.4)	0 (0)	
Bicuspid aortic valve	0 (0)	1 (2.4)	0 (0)	
Epstein anomaly	0 (0)	0 (0)	1 (9.1)	
Double outlet right ventricle	0 (0)	2 (4.8)	0 (0)	
AV canal defect	0 (0)	1 (2.4)	1 (9.1)	
Congenital aortic stenosis	0 (0)	1 (2.4)	0 (0)	
Fontan/single ventricle physiology	**1 (20)**	**15 (35.7)**	**3 (27.3)**	
Comorbidities (%)				
Hypertension	1 (20)	7 (16.7)	1 (9.1)	.775
Coronary artery disease	0 (0)	1 (2.4)	0 (0)	.722
Heart failure	5 (100)	42 (100)	11 (100)	0
Diabetes	1 (20)	3 (7.1)	2 (18.2)	.465
Cerebrovascular accident	0 (0)	10 (23.8)	2 (18.2)	.272
COPD	0 (0)	0 (0)	0 (0)	0
Pulmonary hypertension	2 (40)	22 (52.4)	7 (63.6)	.667
Tobacco history	3 (60)	7 (16.7)	3 (27.3)	.119
Portal hypertension	2 (40)	14 (33.3)	7 (63.6)	.193
Peripheral artery disease	1 (20)	0 (0)	0 (0)	.078
Chronic renal insufficiency	1 (20)	4 (9.5)	3 (27.3)	.329

BMI, body mass index; CHD, congenital heart disease; COPD, chronic obstructive pulmonary disease; NYHA, New York Heart Association.

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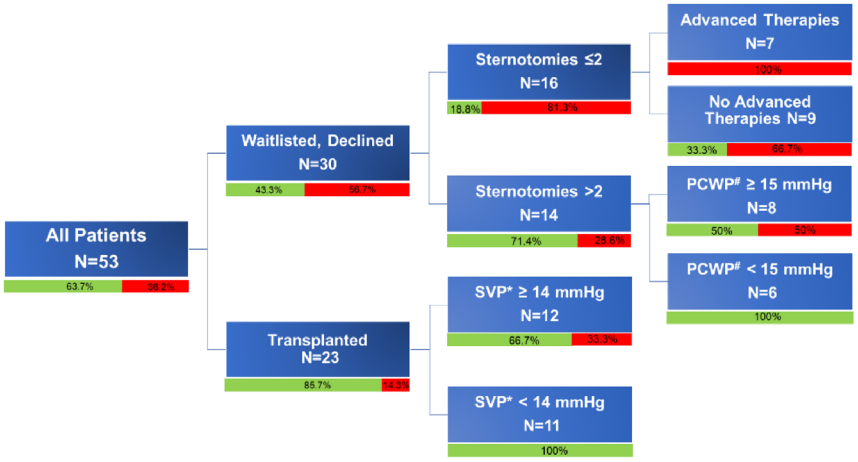
(A)

Parameters for mortality	1 yr-mortality HR (95% confidence interval)	P Value Model for 1-yr mortality	10 yr-mortality HR (95% confidence interval)	P Value Model for 10 yr-mortality
Listing/Transplant status	3.42 [.918 - 12.74]	*p = 0.0022**		*p <.0001**
Declined vs Transplanted	5.15 [1.65 – 16.04]]	*p =0.067**	4.44 [1.64 – 12.0]	*p = .003**
Waitlisted vs Transplanted	N/A (nonsignificant)	*p =0.0047**	5.44 [2.20 – 13.4]	*p = .0002**
# Organs listed (heart vs heart/lung vs heart/liver)		p = 0.706	N/A (nonsignificant)	p = .574
Advanced support	1.90 [.805 – 4.47]			
Yes vs. no		p = 0.148	1.25 [.614 – 2.54]	p = .543
ACHD Complexity	2.11 [.873 – 5.10]			
Moderate vs. Severe Complexity	2.05 [.751 – 5.62]	p = 0.111	2.07 [1.02 – 4.22]	*p = 0.044**
# Sternotomies (≤2) vs. >2	1.05 [.995 -1.107]	p =0.138	1.06 [.525 – 2.13]	p = .875
SVP* (mmHg)	1.08 [1.03 – 1.13]	p = 0.073	1.05 [1.011 – 1.106]	*p = 0.015**
PCWP# (mmHg)	3.42 [.918 - 12.74]	*p = 0.0025**	1.06 [1.023- 1.112]	*p = 0.0018**

(B)



(C)

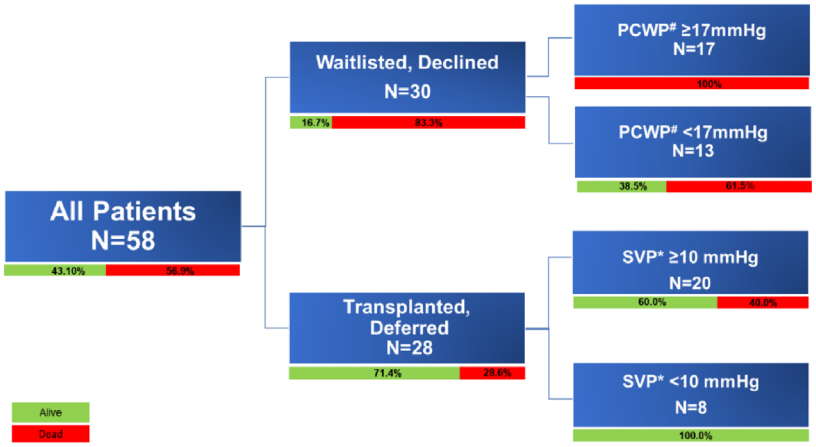


FIGURE 1 Factors affecting 1- and 10-year survival in all adult congenital heart disease (ACHD) patients considered for transplant (A) adjusted hazard rations identify parameters that correlate with 1-year outcomes when patients are transplanted. (B) Decision tree analysis of parameters and their values that correlate with 1-year and (C) 10-year outcomes. *SVP represents pre-transplant systemic venous pressures, #PCWP represents pulmonary capillary wedge pressure.

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TABLE 2 Factors influencing decision for transplantation at the time of evaluation by a multidisciplinary team.

	Deferred/too well *N* (%)	Listed for transplant *N* (%)	Declined/too sick *N* (%)	*p*-value
Number of patients	5 (8)	42 (72)	11 (19)	
# Sternotomies	1 [0,1.5]	2 [1,3]	4 [2,4]	*.0012**
Cather-based hemodynamic testing				
Pulmonary capillary wedge pressure (PCWP) (mmHg)	12.0 [6, 13]	15.0 [12, 21.75]	25.0 [15, 33]	*.0132**
Systemic venous pressure (SVP) (mmHg)	10.0 [7.5, 11.5]	14.5 [8.5, 19]	23.0 [18, 25]	*.0052**
Systemic ventricular end diastolic pressure (mmHg)	12.0 [6,13]	15.0 [12, 21.5]	25.0 [15, 33]	*.0132**
Cardiopulmonary exercise testing				
Peak MVO2 (mL/min/kg)	24 [17.95, 28.1]	15 [12.35, 17.4]	11.7 [9.6, 17.1]	*.0354**
FEV1/FVC median [25, 75]	68 [62.5, 84]	80 [76, 83]	71 [67, 77]	*.0136**
Clinical factors				
Systolic blood pressure (mmHg)	125 [106, 129]	99 [90.5, 110]	102 [98, 110]	*.0178**
Oxygen saturation (%)	95 [86.5, 87.5]	95 [89, 96]	95 [86, 97]	.908
Advanced heart-failure therapies				
Inotropes	1 (20)	16 (38.1)	4 (36.4)	.709
Ventricular assist devices	0 (0)	5 (11.9)	0(0)	–
Extracorporeal membrane oxygenation	1 (20)	1 (2.4)	0(0)	.230
Laboratory values				
Albumin (g/dL)	4.3 [3.9, 4.55]	3.8 [3.1, 4.5]	4.0 [2.9, 4.1]	.357
Creatinine (mg/dL)	.9 [.8, 1.2]	.9 [.8, 1.2]	.9 [.7, 1.5]	.948
Sodium (mEq/L)	138 [137, 139.5]	136 [132.75, 139]	134 [32, 138]	.1823
BNP (pg/mL)	156 [43, 287]	293 [140.5, 1270.5]	614 [63, 1234]	.329
Complexity based on ACHD AP classification	.0628			
Moderate	2 (40)	8 (19)	6 (54.5)	
Severe	3 (60)	34 (81)	5 (45.5)	

ACHD, adult congenital heart disease; MVO2, myocardial oxygen consumption. p

3.4 | Days to transplantation did not affect 1-year survival

Days to transplantation after decision to list was not associated with 1 year survival outcomes. Qualitative analysis of our data showed that heart + lung transplantation had wait periods of 220 days whereas heart transplantation alone had a median wait period of 117 days (Table S2A). Based on blood type wait periods ranged between 50 and 479 days (Table S2B).

3.5 | Ten-year survival outcomes correlated with invasive hemodynamic parameters

Transplantation also correlated strongly with 10-year survival in CHD patients, with unadjusted analysis showing that transplanted patients were 5.44 times more likely to survive compared to waitlisted patients, and 4.44 times more likely to survive compared to patients declined for transplantation (Table 1A). Multivariate decision tree analysis (Table 1C) indicated that 100% of patients who were waitlisted or declined with PWCP ≥17 mmHg had died by 10-years. Similarly, 100% of transplanted or deferred patients with SVP < 10 mmHg were alive at 10 years. Transplanted patients with SVP ≥10 mmHg had a 60% survival, whereas waitlisted/declined patients with PCWP > 17 mmHg only had a 38.5% survival rate.

Kaplan–Meier survival analysis (Figure 2) of patients considered for transplantation showed a significant difference in 10-year survival between patient cohorts.

3.6 | Kaplan–Meier analysis identified discrepancies in survival outcomes based on listing decision

Kaplan–Meier survival analysis identified outcomes in ACHD patients binned into four categories: deferred/too well, waitlisted (not transplanted), transplanted, or declined/too sick (Figure 2). We found that the deferred population had the greatest 1-year (100%) and 10-year survival (80%). The waitlisted group had the lowest 1- and 10-year survival (42.1% and 15.9%, respectively). Patients declined due to being too sick for transplantation had the lowest 30-day survival at 81.8%

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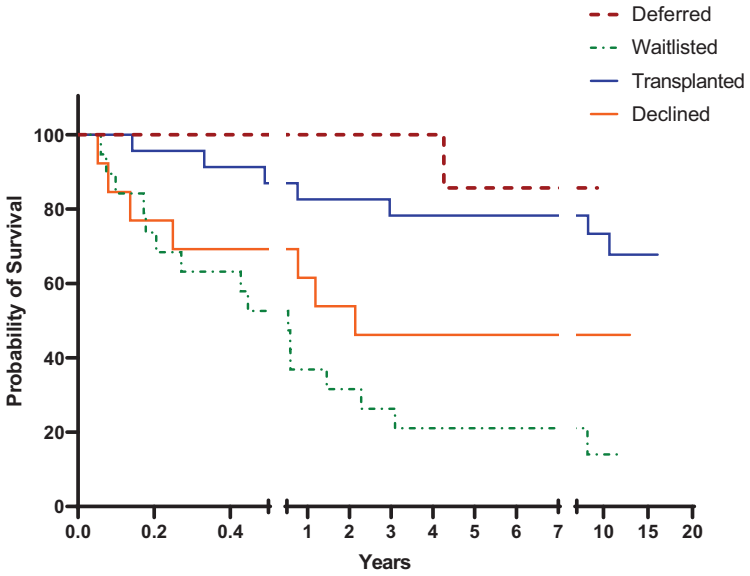

Group	0d	30d	1 y	5 y	10 y
Deferred	5 (100)	5 (100)	5 (100)	5 (100)	4 (80)
Transplanted	23 (100)	20 (87.0)	19 (85.7)	16 (69.6)	16 (69.6)
Waitlisted	19 (100)	17 (89.5)	8 (42.1)	3 (15.9)	3 (15.9)
Declined	11 (100)	9 (81.8)	6 (54.5)	2 (18.2)	2 (18.2)

FIGURE 2 Kaplan–Meier survival curves for 10-year survival from date of decision show a highest 30-day mortality in patient declined (too sick) for transplantation, and highest 1- and 10-year mortality in waitlisted patients. While deferred (too well) patients had the highest survival a 1- and 10-year, transplanted patients had improved survival as compared to patients who were not-transplanted.

and the one of the lowest survival at 10-year at 18.2%. Transplanted patients had a significant improvement in survival as compared to the waitlisted and declined groups throughout the study period (30 days = 87%, 1 year = 85.7%, 5 years = 69.6%, 10 years = 69.6%). Small patient populations prohibited identification of additional factors helping with survival in waitlisted or declined patients.

3.7 | Outcomes in single ventricle patients

Of the 19 single ventricle patients, one year survival irrespective of status was 79.3% versus 59.0% for all other CHD diagnoses (p = .265). Ten-year survival was 57.9% for single ventricle patients versus 35.9% for other CHD diagnoses (p = .116). None of the analyzed factors (PVR, Fick index, wedge pressure, median sternotomies, age at eval, SVP, inotropes, VO2) correlated with 1- or 10-year mortality.

4 | DISCUSSION

Our study evaluated a comprehensive list of hemodynamic, laboratory, and functional parameters to identify factors that correlate with outcomes in ACHD patients at 1- and 10-year post-decision for transplantation. Transplantation improves survival at 1 and 10 years in ACHD patients. In our cohort, 1-year survival improved 5.15-fold and 10-year survival improved 5.44-fold in ACHD patients who were transplanted as compared to those who were waitlisted and never received transplant.

4.1 | Cardiac anatomic predisposition to transplantation

D-TGA and atrial switch operation was the most common CHD diagnosis in our cohort (Table 1). These patients are by far the oldest in the group and have an extensive history of arrhythmia and heart failure (100%) resulting from a systemic right ventricle. As atrial switch operations have been replaced with arterial switch operation, where the left ventricle is systemic, ACHD patients with D-TGA/atrial switch undergoing transplantation is anticipated to decrease in the upcoming years. In the coming decades, this population will be replaced by patients with tetralogy of Fallot, tricuspid atresia (with Fontan physiology) and cc-TGA as the anatomic group most in need of transplantation. TOF is the most common cyanotic CHD and hence reflects the large number of patients with heart failure in need of transplantation. The long-term outcome of repaired TOF patients is largely reassuring with high survival rates, however in patients with significant pulmonary regurgitation, RV dilation, ventricular arrhythmia, biventricular failure and a higher risk of sudden cardiac death[7,8] is likely. The optimal timing for pulmonary valve replacement in these patients is not well understood and likely contributes to their need for transplantation.

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Clinical TRANSPLANTATION
The Journal of Clinical and Translational Research
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Tricuspid atresia is the most common diagnosis in adults with univentricular physiology.[9] Patient with Fontan circulation pose unique challenges with transplantation with additional anatomic risk factors including presence of collaterals, pulmonary artery anatomy, and additional comorbidities including PLE contributing to challenges in transplantation. Given the varied anatomies of single ventricle patients, this study was not sufficiently powered to identify factors influencing post-transplant outcomes in single ventricle patients.

4.2 | Factors affecting listing parameters and their relationship to survival outcomes

Invasive hemodynamic testing and exercise testing primarily correlated with decision to list. Our model validated criteria that identified patients who were too well to undergo transplantation as they had 100% 1-year survival and 80% 10-year survival. Numerous laboratory parameters including renal function and albumin that influence listing in traditional patients were found to be near normal in our ACHD patients (Table 2). Listing decisions by our multidisciplinary transplant listing team correlated with sternotomies, hemodynamic parameters, and CPET and correlated with outcomes at 1- and 10-years post-decision for transplantation.

4.3 | More median sternotomies do not correlate with CHD complexity and outcomes

Surprisingly, patients with ≤2 sternotomies were more likely to die (81.3%) while waitlisted or declined patients - had 100% mortality when placed on advanced therapies. Of these patients 43.8% had a systemic right ventricle either due to atrial switch operation or congenitally corrected-TGA. Multiple factors contribute to this mortality including echocardiographic limitations associated with imaging right ventricles, high predisposition to lethal arrhythmias, and subsequent limitations with using advanced imaging techniques such as cardiac magnetic resonance (cMR) due to presence of devices such as pacemakers and defibrillators. Furthermore, the efficacy of traditional heart failure therapies has limited translation in systemic right ventricles. Additionally, most patients in this class have few functional limitations as children and were discharged from pediatric cardiology services without transition to ACHD providers. Another 31% of patients were single ventricle patients with limited palliation. These patients were notably cyanotic with saturations < 85% and also had pulmonary hypertension as their comorbidities. Both category of patients are the sickest patients in this group with 100% mortality. When advanced heart-failure therapies are not needed their survival improves to 33.3% at 1 year.

Patient with >2 median sternotomies and wedge pressures <15 mmHg have a 100% survival rate at 1 year. Patients in this group have a diagnosis of tetralogy of Fallot or double outlet right ventricle and have a history of surgical revision of conduits and valves. All transplanted patients with SVP < 14 mmHg at the time of listing had a 100% survival at 1-year. These results underscore the importance of continued surveillance of CHD in adulthood with multimodality diagnostic testing with invasive hemodynamic catheterization and CPET.

4.4 | Triaging listing status correlates with survival outcomes

As compared to other published studies, a larger percentage of patient were transplanted at our center.[10] Crossland et al. represented a European cohort where only 1/3rd of evaluated patients were transplanted. One year outcomes were significantly better as compared to outcomes from the thoracic ISHLT registry which reported a 78.3% survival and comparable to outcomes in patients with ischemic (84.3%) and dilated cardiomyopathy (86.2%).[11] Kaplan–Meier survival analysis demonstrated that our post-transplantation group had improved 10-year mortality (69.6%) compared to 59.3% reported by other groups.[12] Interestingly, our results showed worse survival (42.1%) in the waitlisted population who awaited transplantation, compared to other studies looking at a similar population which reported a 1-year survival of 89%, 63% at 24 months, and 63% at 48 months.[13] These data may reflect a sicker patient population listed for transplantation at our center that is largely driven by single ventricle patients with limited palliation and systemic right ventricle patients with delayed identification for transplantation and late referrals from other centers. As expected, deferred patients who are too well for transplantation have better survival than waitlisted patients; however, patients declined transplantation also have better survival than waitlisted patients. The improved survival in declined patients could be secondary to other criteria such as high allosensitization and social and financial limitations which deems them to be poor transplant candidates.

4.5 | One- and 10-year survival in patients largely correlates with intracardiac pressures

Various clinical comorbidities have been shown to predict prognosis in ACHD populations including pulmonary hypertension, restrictive lung disease, anemia, and renal dysfunction.[14–17] Exercise capacity testing has also revealed that peak VO2 and VE/VCO2 have been shown to be independent predictors of mortality in ACHD patients.[18,19] While these parameters may help grossly adjust risk quantification, ACHD patients are an extremely heterogenous group and additional parameters are needed to quantify patient specific risk. Right heart catheterization hemodynamic measurements may reflect more individualized parameters to assess short- and long-term prognosis. Unfortunately, there has been a lack of studies specifically analyzing these parameters.

One-year mortality at our center matched national statistics after transplantation in ACHD patients, despite a significantly higher

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population of single ventricle patients. Listing for heart versus heart +additional organs did not correlate with outcomes. Studies from other centers indicate a multi-organ transplant with increased mortality.[20] Instead, ACHD complexity, SVP, and PCWP all were significant predictors of 1 year survival in ACHD patients. In addition, both SVP and wedge pressure were significant predictors of 10-year mortality.

4.6 | Study limitation

The small sample size of 58 patients with heterogeneity of diagnosis is a limitation of this study. These parameters must be further validated in larger cohorts. The retrospective nature of our study also limits our ability to validate any prognostic markers. As a single center transplant center, our outcomes may not be reflective of patients presenting at other centers. The study evaluated transplants between 2004 and 2014, which precedes the current UNOS transplantation listing criteria. Hence, intervals between listing and transplantation may be different in the current era. However, as we continue to evaluate the same clinical parameters, the findings of this study are relevant to triaging patients for transplantation. Due to a longer era of atrial switch operations in pediatric patients at our center, we have a higher incidence of patients with Senning/Mustard operations.

5 | CONCLUSION

We have identified hemodynamic, anatomic, and clinical factors that are associated with both 1- and 10-year survival in our adult patients with CHD, which correlate with improved survival in ACHD patients at our center.

ACKNOWLEDGMENTS

This study was funded by the National Institutes of Health K08 (K08 HL 161440), American Heart Association Career Development Award (852875), and Foundation grant from HeartFest to A.S.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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SUPPORTING INFORMATION

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How to cite this article: Hickey G, Ratnayake C, Elliot A, et al. Long-Term outcomes in adult patients with congenital heart disease considered for transplantation: A single center study. *Clin Transplant*. 2023;37:e15101. https://doi.org/10.1111/ctr.15101

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